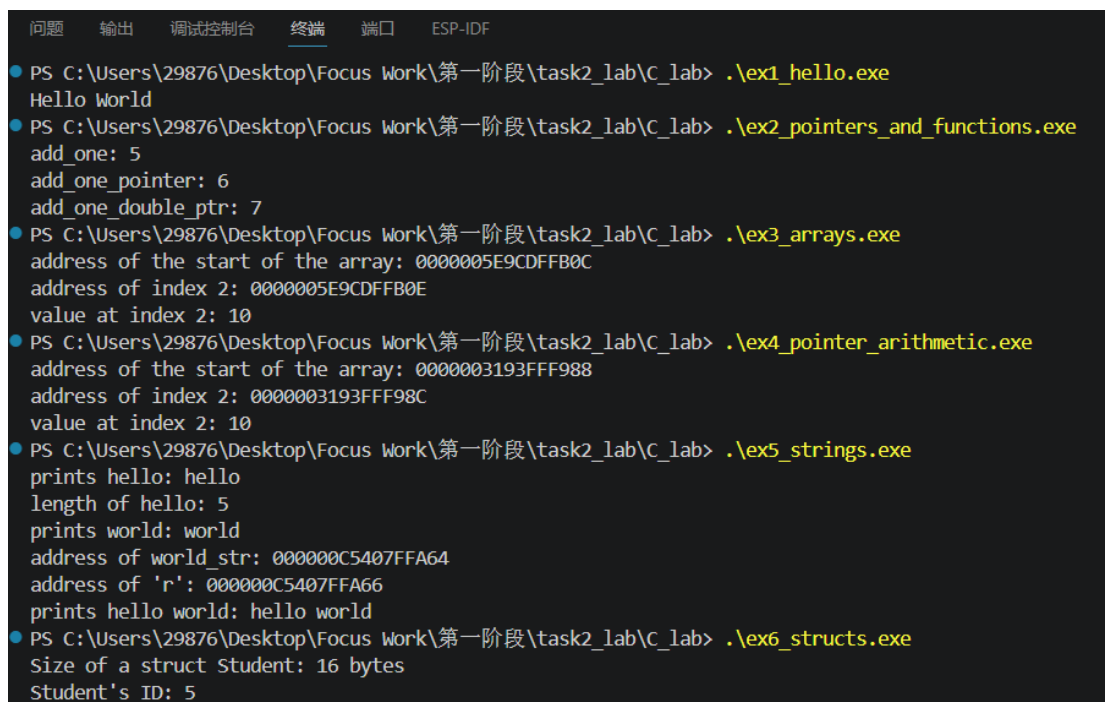


# 第一阶段报告

完成情况：完成了 C\_lab 和 Python\_中的所有内容

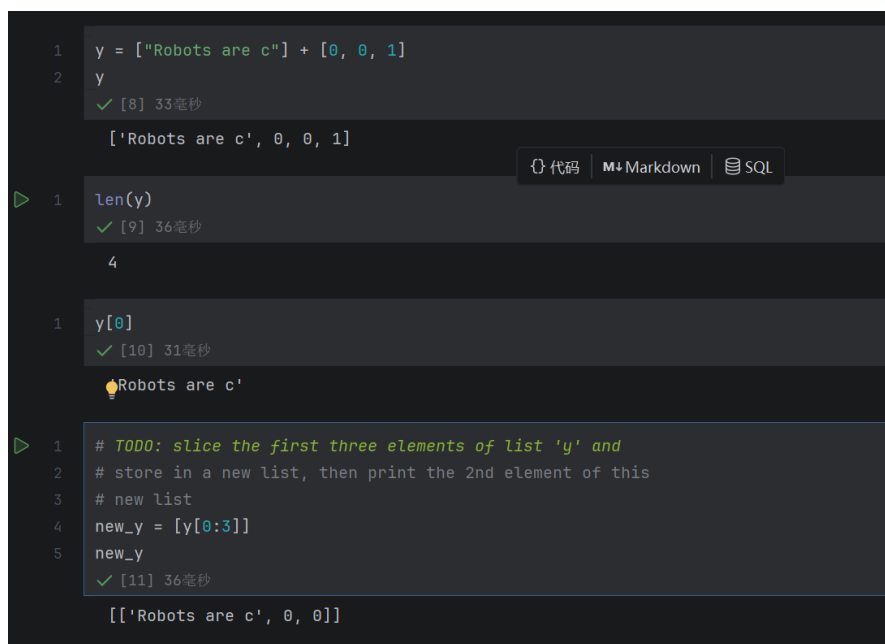
C\_lab 运行情况截图：



```
问题 输出 调试控制台 终端 端口 ESP-IDF
● PS C:\Users\29876\Desktop\Focus Work\第一阶段\task2_lab\C_lab> .\ex1_hello.exe
Hello World
● PS C:\Users\29876\Desktop\Focus Work\第一阶段\task2_lab\C_lab> .\ex2_pointers_and_functions.exe
add_one: 5
add_one_pointer: 6
add_one_double_ptr: 7
● PS C:\Users\29876\Desktop\Focus Work\第一阶段\task2_lab\C_lab> .\ex3_arrays.exe
address of the start of the array: 0000005E9CDFFB0C
address of index 2: 0000005E9CDFFB0E
value at index 2: 10
● PS C:\Users\29876\Desktop\Focus Work\第一阶段\task2_lab\C_lab> .\ex4_pointer_arithmetic.exe
address of the start of the array: 0000003193FFF988
address of index 2: 0000003193FFF98C
value at index 2: 10
● PS C:\Users\29876\Desktop\Focus Work\第一阶段\task2_lab\C_lab> .\ex5_strings.exe
prints hello: hello
length of hello: 5
prints world: world
address of world_str: 000000C5407FFA64
address of 'r': 000000C5407FFA66
prints hello world: hello world
● PS C:\Users\29876\Desktop\Focus Work\第一阶段\task2_lab\C_lab> .\ex6_structs.exe
Size of a struct Student: 16 bytes
Student's ID: 5
```

Python\_lab 运行情况截图：

(1) Lists 部分的 todo:



```
1 y = ["Robots are c"] + [0, 0, 1]
2 y
✓ [8] 33毫秒
['Robots are c', 0, 0, 1]
代码 Markdown SQL
▶ 1 len(y)
✓ [9] 36毫秒
4
1 y[0]
✓ [10] 31毫秒
🔦 Robots are c'
▶ 1 # TODO: slice the first three elements of list 'y' and
2 # store in a new list, then print the 2nd element of this
3 # new list
4 new_y = [y[0:3]]
5 new_y
✓ [11] 36毫秒
[['Robots are c', 0, 0]]
```

(2) LOOPS 部分的 todo:

You can loop over the elements of a list like this:

```
1 robots = ['baxter', 'sawyer', 'turtlebot']
2 for robot in robots:
3     print(robot)
4 # Prints "baxter", "sawyer", "turtlebot", each on its own line.
✓ [12] 63毫秒
```

```
baxter
sawyer
turtlebot
```

If you want access to the index of each element within the body of a loop, use the built-in `enumerate` function.

```
1 robots = ['baxter', 'sawyer', 'turtlebot']
2
3 # TODO: Using a for loop and the python built-in enumerate function,
4 # Print "#1: baxter", "#2: sawyer", "#3: turtlebot",
5 # each on its own line
6 for index,robot in enumerate(robots):
7     print(f"#1: {robot}")
✓ [13] 33毫秒
```

```
#0: baxter
#1: sawyer
#2: turtlebot
```

(3) slicing numpy arrays 部分的 todo:

```
1 A = np.array([[5, 6, 8], [2, 4, 5], [3, 1, 10]])
2 B = np.array([[3, 5, 0], [3, 1, 1]])
3 # TODO: multiply matrix A with matrix B padded with 1's to the
4 # same dimensions as A; sum this result with the identity matrix
5 # (you may find np.concatenate, np.vstack, np.hstack, or np.eye useful).
6 # Make sure you don't alter the original contents of B. Print the result
7 one_array = np.ones((1,3),dtype=int)
8 copy_B = np.copy(B)
9 C = np.vstack((copy_B,one_array))
10 D = np.dot(A,C)
11 result = D + np.eye(3)
12 print(result)
✓ [18] 38毫秒

[[42. 39. 14.]
 [23. 20.  9.]
 [22. 26. 12.]]
```

(4) Handy Numpy function `:arange` 部分的 todo:

```
[-3 -2 -1  0  1  2  3]

# TODO: print every other integer from 0 ... 6 multiplied by 2
# as a list
print(np.arange(0,7,2)*2)
✓ [17] 34毫秒

[ 0  4  8 12]
```

Python\_lab 扩展部分:

```
PS C:\Users\29876\Desktop\Focus Work\第一阶段\task2_lab\python_lab\hog> python ok
=====
Assignment: Hog Project
OK, version v1.18.1
=====

~~~~~
Running tests

<input>:1: SyntaxWarning: 'bool' object is not callable; perhaps you missed a comma?
<input>:1: SyntaxWarning: 'bool' object is not callable; perhaps you missed a comma?
-----
Test summary
    267 test cases passed! No cases failed.

Backup... 100% complete
Backup past deadline by 380 days, 22 hours, 54 minutes, and 5 seconds
Backup successful for user: ghy070308@gmail.com
URL: https://okpy.org/cal/cs61a/fa25/hog/backups/BQ5ymY

OK is up to date
```

```
-----
Point breakdown
    Question 0: 0.0/0
    Question 1: 2.0/2
    Question 2: 2.0/2
    Question 3: 2.0/2
    Question 4: 2.0/2
    Question 5: 4.0/4
    Question 6: 2.0/2
    Question 7: 2.0/2
    Question 8: 2.0/2
    Question 9: 2.0/2
    Question 10: 2.0/2
    Question 11: 2.0/2
    Question 12: 0.0/0

Score:
    Total: 24.0

Backup... 100% complete
Backup past deadline by 380 days, 22 hours, 54 minutes, and 45 seconds
Backup successful for user: ghy070308@gmail.com
URL: https://okpy.org/cal/cs61a/fa25/hog/backups/DP5AoK

OK is up to date
```